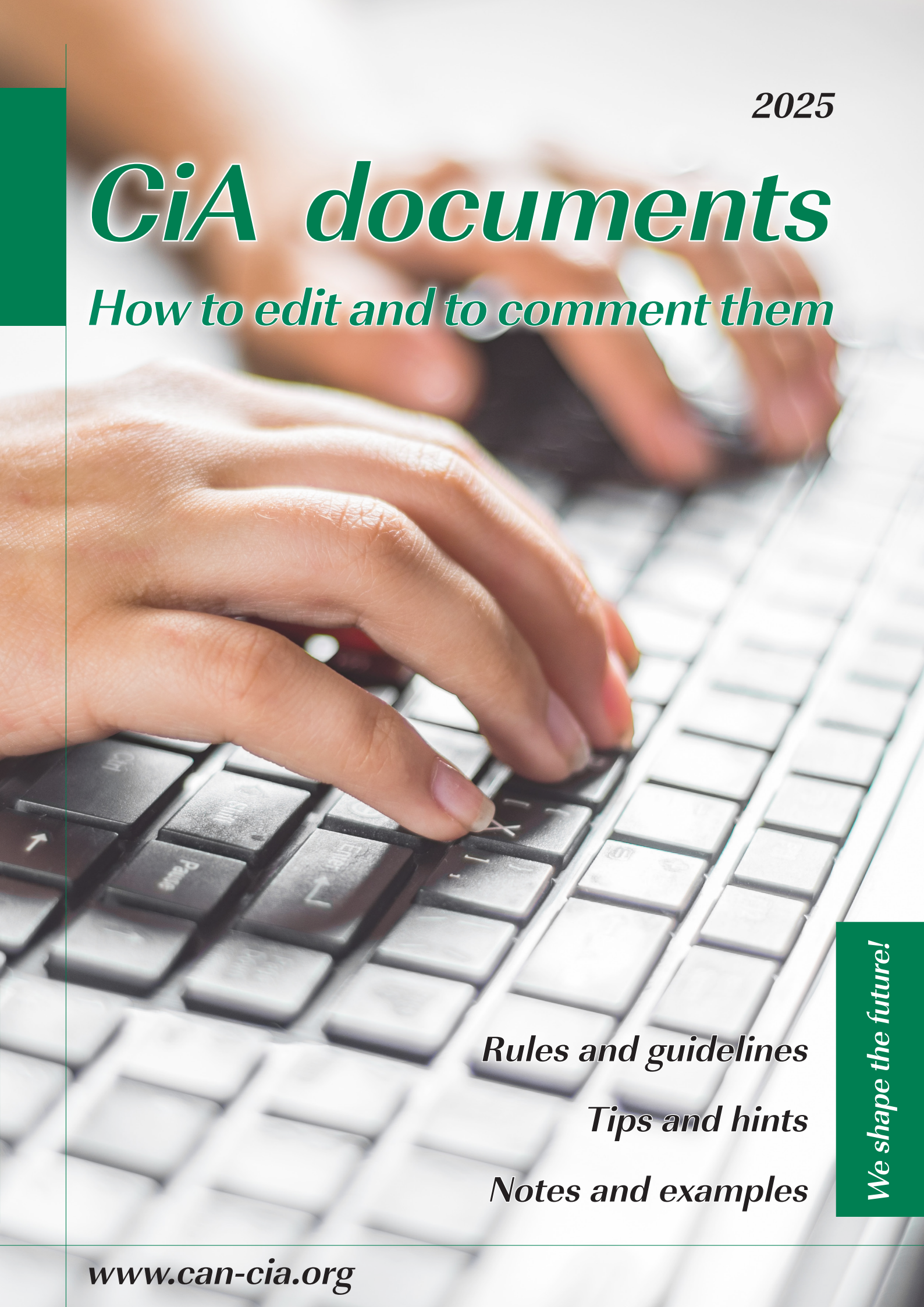
2025

CiA documents

How to edit and to comment them



Rules and guidelines

Tips and hints

Notes and examples

We shape the future!

www.can-cia.org

How this brochure can help you

This brochure is designed to help people, writing clear, concise, and user-friendly CiA technical documents. In general, CiA has adapted the ISO/IEC Directives, Part 2 (Principles and rules for the structure and drafting of ISO and IEC documents), providing drafting and editorial rules. Some more detailed editorial guidelines are given in the CiA house style document.

Writing plain language

Write CiA technical documents with the reader in mind, who is eventually not an English native speaker. Using plain language is an effective means of getting your message across. By being clear and concise – but not simplistic – writers can avoid misinterpretation. This reduces time and cost in translation. Because plain language is easier to understand, it also reduces discussions during drafting. Using plain language does not mean reducing the length of your message, changing its meaning or oversimplifying your text.

How to use plain language

- Be clear to yourself about your main message – try reading it to yourself out loud
- Put yourself in the place of the reader
- Keep your sentences short
- Have one idea per sentence
- Leave out words you don't need
- Use lists when you can
- Use the active voice when you can
- Be concise, use short, simple words and avoid turning verbs into nouns
- Punctuate your writing carefully
- Use more full stops, fewer commas and brackets
- Phrase your points positively
- Use everyday language whenever possible and reduce jargon
- Plain language is particularly important in the Scope

Hint

Use the same term for the same concept everywhere. Don't use synonyms and don't use one term for different concepts!

Definition or specification

A definition (to define) explains the meaning of something. A specification (to specify) determines details of something, e.g., features. They are not synonyms.

Verbal forms

In the whole CiA technical document, be clear about what is a requirement and what is a recommendation or other type of statement. In order to make clear what the implementor must do, the following verbal forms are used in CiA technical documents:

- Requirement: shall/shall not
- Recommendation: should/should not
- Permission: may
- Possibility and capability: can/cannot
- External constraint: must/must not

CiA technical reports do not contain requirements, but can contain recommendations and permissions.

Hint

Avoid double-specifications: Do not duplicate requirements, recommendations, and permissions (e.g., in a table and in a figure).

Document structure

CiA technical documents comprise the following items:

- Cover page with title
- Foreword section
- Introduction section (optional)
- Main body section (clauses and subclauses)
- Annexes (optional)
- Bibliography section (optional)

Document numbering and status

CiA numbers uniquely technical documents. They are classified as CiA specifications or CiA technical reports (TR), when they are released. CiA specifications can be in Draft Specification Proposal (DSP), Draft Specification (DS), or in Public Available Specification (PAS) status.

Title

The document title must be clear and concise, containing a maximum of two elements:

- a main element
- an optional complementary element, indicating the subtitle of a part

Example

CiA 306-1:
CANopen electronic device description – Part 1: Electronic data sheet (EDS) and device configuration file (DCF)

CiA 406-C:
Encoder device profile – Part C: Mapping to CANopen CC

Foreword

The Foreword section contains generic text, which is inserted by CiA office during editing and publishing. When revising an existing CiA document, include a list of the major changes compared with the previous edition.

Introduction

The Introduction section is optional but CiA encourages its inclusion. It can describe the content of a document series and give background information on why it is needed.

Example

CiA 408 series:
This set of CiA specifications comprises the following documents:

- *CiA 408-B: Fluid power technology device profile — Part B: Functional device description and parameter;*
- *CiA 408-C: Fluid power technology device profile — Part C: Mapping to CANopen CC;*
- *CiA 408-F: Fluid power technology device profile — Part F: Mapping to CANopen FD.*

This set specifies the functionality of interconnectable proportional valves, hydrostatic pumps, and hydrostatic transmissions with mappings to CANopen CC and CANopen FD technologies. The set is based on the profile "Fluid Power Technology", version 1. 7.0, designed by Verband Deutscher Maschinen- und Anlagenbau e.V. Frankfurt/Main, Germany (VDMA).

Table of contents

The table of contents is automatically generated.

Scope

The Scope clause is mandatory and it describes what the document does. For example, this document

- specifies ...
- establishes ...
- gives guidelines for ...
- defines terms ...

The Scope clause is written as a series of statements of fact. Don't put any requirements, recommendations, or permissions in the Scope.

Hint

Check that information is not duplicated between the Introduction section and the Scope clause!

Example

CiA 408-B:
This document specifies functional principles and application parameters for hydraulic proportional valves, hydrostatic pumps, and hydrostatic transmissions. They can be applied on pneumatic, as well.

Normative references

The Normative references clause is mandatory, even if there are no normative references in the document. It lists reference documents, which are cited in the text in such a way that some or all of their content constitutes requirements of the document. Remember to date the reference, if it refers to a specific clause, sub-clause, figure, table, etc. References are generally made to CiA documents. Documents from other organizations can also be referenced; references must be publicly available.

The Normative references clause has an introduction sentence: "The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies." Consider that (sub-) clause, figure, and table numbers can be different in released versions.

Terms and definitions

The Terms and definitions clause is mandatory and defines the meaning of certain terms in the context of the document. Only terms, which are used in the document are defined.

A definition is a single phrase that can replace the term wherever used. It does not start with an article (e.g. " a ", " the ") or end with a full-stop. It does not take the form of, or contain, a requirement or recommendation. Additional information can be included in a Note to entry or an example.

The Terms and definitions clause has an introduction sentence: "For the purposes of this document, the terms and definitions given in the CiA terminology database (https://www.can-cia.org/s/glossary-of-terms) and the following apply."
The Terms and definitions clause has also a last sentence: "Additionally, ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform (https://www.iso.org/obp/)
- IEC Electropedia (https://www.electropedia.org/)"

Example

CiA 1301:

data element

single element of a data object, identified uniquely, by a 16-bit index and an 8-bit sub-index

Symbols and variables

Abbreviated terms, symbols, and variables may be listed in a separate clause after the Terms and definitions clause.

Variables use only one letter. They are written in Italic font; optional subscripts use upright font. Time variables use *t*, and temperature variables shall use *T*, for example.

Clauses and subclauses

Clauses and subclauses form the main part of any CiA technical document. They tell readers what they need to do to implement it. Number clauses and subclauses to help people reference key parts of the document, do not use "hanging paragraphs", which is a text introduced between (sub-) clause levels.

CiA does not allow its documents to mandate the use of services, such as testing or certification (e.g., by another company). Write the requirements so they can be verified by anyone.

CiA technical documents should avoid referring to trademarks or companies. Patented items can be referred to under certain conditions: For details see the Foreword section and the listed patents on the CiA website. CiA technical documents do not contain legal or contractual requirements.

Hint

Subclauses can be with or without headings. But be consistent within a subclause.

Notes and examples

Notes and examples are used for giving additional information intended to assist the understanding or use of the document. Don't put any requirements, recommendations, or permissions in the Notes or examples.

Hint

If the Note is removed, do you lose essential information? If so, it should not be a Note.

Hint

Table cells should not be empty; use "not applicable (n.a.)", for example.

Tables and figures

Cite all tables and figures:

- Normative citation: "Table/Figure X specifies/defines .."
- Informative citation: Table/Figure Y shows/provides .."

Place the table/figure reference in the clause or the subclause, where they are placed.

Annexes

Annexes are used to provide additional information. They can be normative (e.g., a test method that the user is required to follow) or informative (additional information that complements the user's understanding). The status (normative or informative) is determined by how the annex is cited in the main body of the document. Annexes are designated by a capital letter (A, B, C, ..). Normative annexes shall come first.

Bibliography

List documents in the Bibliography section that provide background information. Be sure to list all references cited in the document either in the Bibliography or in Clause 2 (depending on how they are cited in the document).

Hint

Keep the Bibliography section as concise as possible.

Additional information

For more information, please visit www.can-cia.org:

- List of released CiA technical documents
- Development process of CiA technical documents
- Glossary of terms
- CANdictionary
- List of referenced patents
- CiA rules and regulations
- CiA antitrust compliance statement
- CiA artificial intelligence (AI) statement

Comment template

The CiA comment template should be used, when submitting comments and additional functional requests.

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